Practical No. 6

Aim: Import the data warehouse data in Microsoft Excel and create the Pivot table and Pivot Chart to perform data analysis.

Create Pivot Table for following:

1) Number of Customers by Gender
   Rows: Gender
   Values: Count of CustomerKey

Row Labels	Count of CustomerKey
F	8892
M	9126
NA	130
Grand Total	**18148**

Explanation: This table shows how many male and female customers are in our data.

2) Average Income by Education Level
   Rows: Educational Level
   Values: Average of AnnualInacome

Row Labels	Average of AnnualIncome
Bachelors	64396.50257
Graduate Degree	66150.4
High School	48963.28294
Partial College	55090.61619
Partial High School	39575.5627
Grand Total	**57269.12056**

Explanation: This shows which group of people earns more based on their school or college degree.

3) Customers by Occupation and Marital Status
   Rows: Occupation
   Column: MartialStatus
   Values: Count of CustomerKey

Count of CustomerKey	Column Labels		
Row Labels	M	S	Grand Total
Clerical	1407	1452	2859
Management	1888	1123	3011
Manual	791	1562	2353
Professional	3062	2362	5424
Skilled Manual	2669	1832	4501
Grand Total	**9817**	**8331**	**18148**

Explanation: This helps us see if certain jobs have more married or single people.

4) Home Owners vs non-Home Owners
Rows: HomeOwner
Values: Count of CustomerKey

Row Labels	Count of CustomerKey
Y	12260
N	5888
Grand Total	**18148**

Explanation: This table tells us how many customers own a house and how many do not.

5) Total Children by Gender
Rows: Gender
Values: Count of TotalChildren

Row Labels	Count of TotalChildren
F	8892
M	9126
NA	130
Grand Total	**18148**

Explanation: This shows the total number of children belonging to male and female customers.

6) Average Income by Gender and Occupation
Rows: Gender, Occupation
Values: Average of AnnualIncome

Row Labels	Average of AnnualIncome
Clerical	**30696.04757**
F	30857.3487
M	30575.9889
NA	29000
Management	**92218.53205**
F	91505.89868
M	92978.03618
NA	85454.54545
Manual	**16472.58819**
F	16153.15315
M	16748.36601
NA	17368.42105
Professional	**74166.66667**
F	73922.28739
M	74446.11341
NA	71818.18182
Skilled Manual	**51732.94823**
F	51968.53933
M	51448.88889
NA	56153.84615
Grand Total	**57269.12056**

Explanation: This shows the average salary for different jobs, split between men and women.

7) Customers by Birth Year
Rows: Decade
Values: Count of CustomerKey

Row Labels	Count of CustomerKey
1910	18
1920	55
1930	627
1940	2199
1950	4155
1960	6100
1970	4741
1980	253
Grand Total	**18148**

Explanation: This groups customers by their age group (like the 70s or 80s) to see which generation is largest.

8) Average Income by Marital Status
Rows: Marial Status
Values: Average of AnnualIncome

Row Labels	Average of AnnualIncome
M	60313.74147
S	53681.4308
Grand Total	**57269.12056**

Explanation: This table shows if married or single people have a higher average salary.

9) Number of Customers by Total Children
Rows: TotalChildren
Values: Count of CustomerKey

Row Labels	Count of CustomerKey
0	5080
1	3552
2	3703
3	2153
4	2259
5	1401
Grand Total	**18148**

Explanation: This shows how many customers have zero, one, two, or more children.

10) Occupation-wise Home Ownership

Rows: Occupation

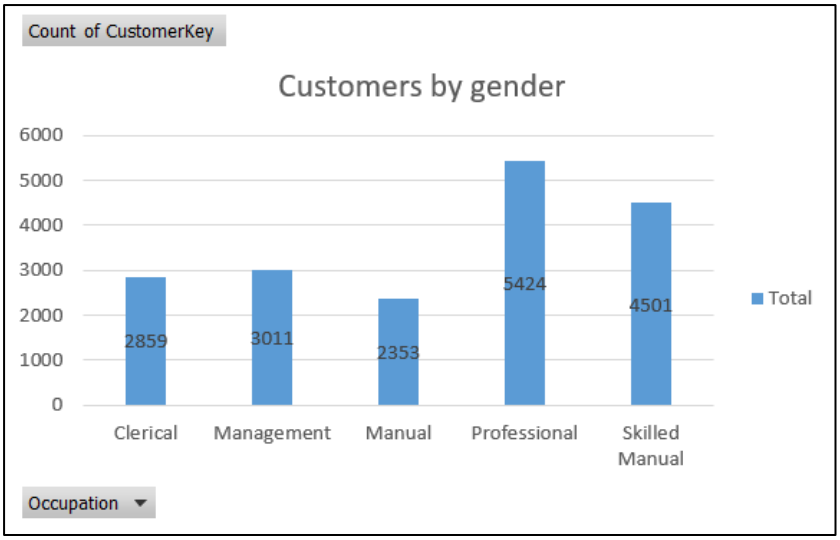
Values: Count of HomeOwner

Row Labels	Count of HomeOwner
Clerical	2859
Management	3011
Manual	2353
Professional	5424
Skilled Manual	4501
Grand Total	**18148**

Explanation: This shows which types of jobs are held by people who own their own homes.

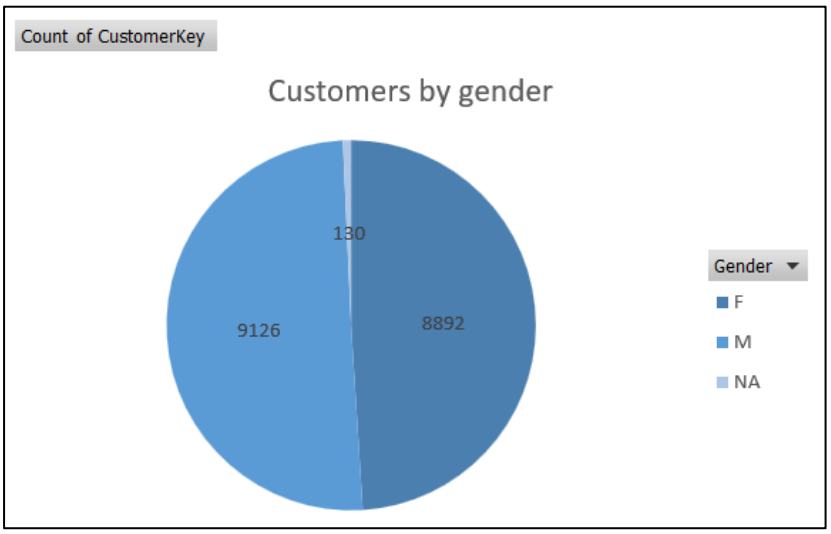
Create Pivot Chart for following

1) Customers by Occupation (use column chart)
   Axis (Categories): Occupation
   Values: Count of CustomerKey



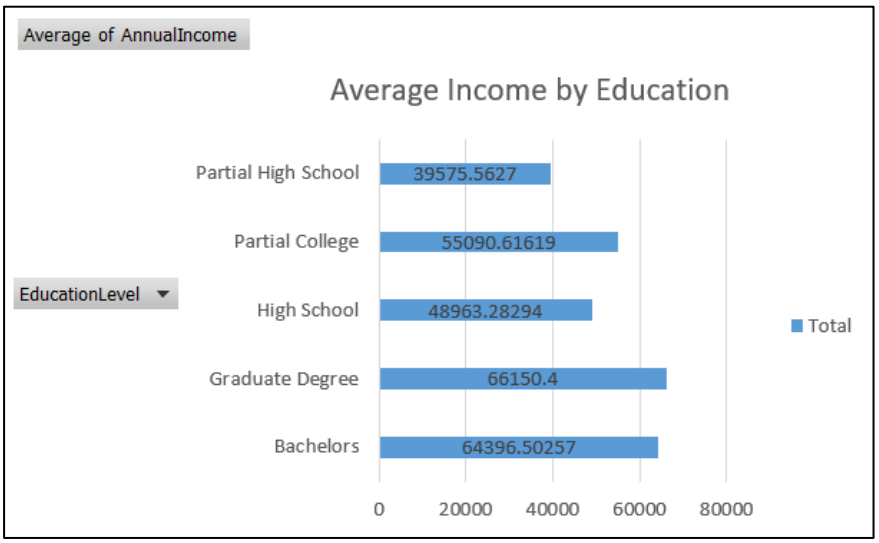
Explanation: A chart with vertical bars that makes it easy to see which job is the most common.

2) Customers by gender (use pie chart)
   Axis (Categories): Gender
   Values: Count of CustomerKey



Explanation: A circle chart that shows the percentage of male vs. female customers.

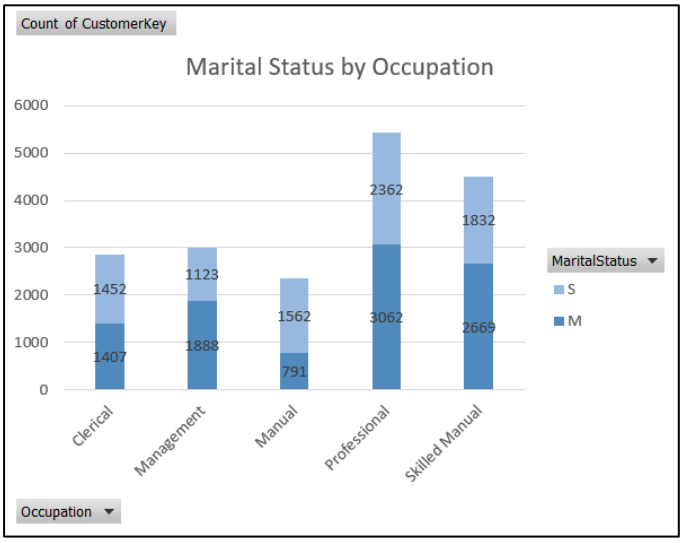
3) Average Income by Education(use bar chart)
Axis (Categories): EducationalLevel
Values: Average AnnualIncome



Explanation: A horizontal chart that ranks education levels from highest to lowest income.

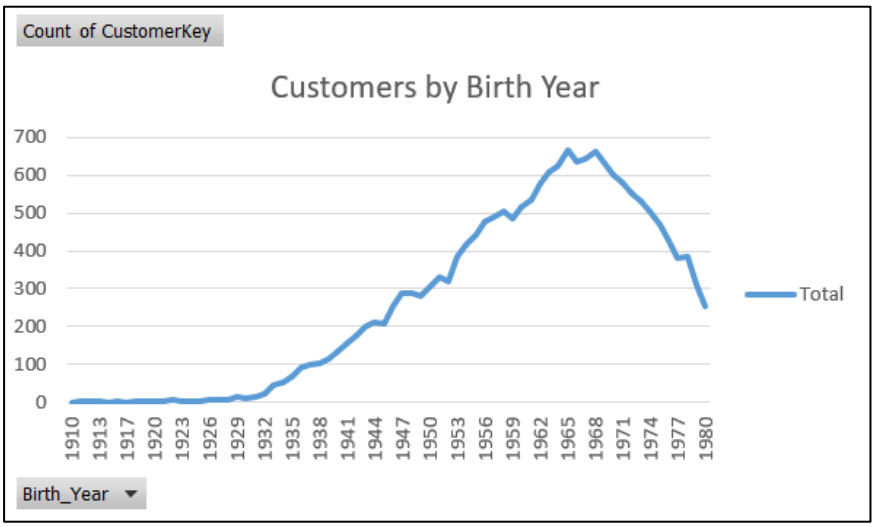
4) Marital Status by Occupation(use Stacked Column Chart)

Legend (Series): MartialStatus
Axis (Categories): Occupation
Values: Count of CustomerKey



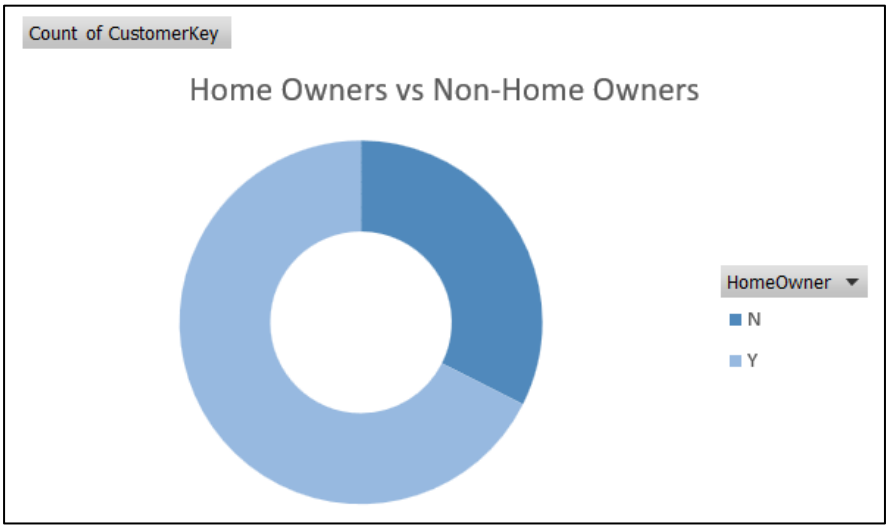
Explanation: One bar per job that is colored to show the mix of married and single people.

5) Customers by Birth Year(Line Chart)
Axis (Categories): Birth_Year
Values: Count of CustomerKey



Explanation: A line that goes up and down to show how many customers were born in each year.

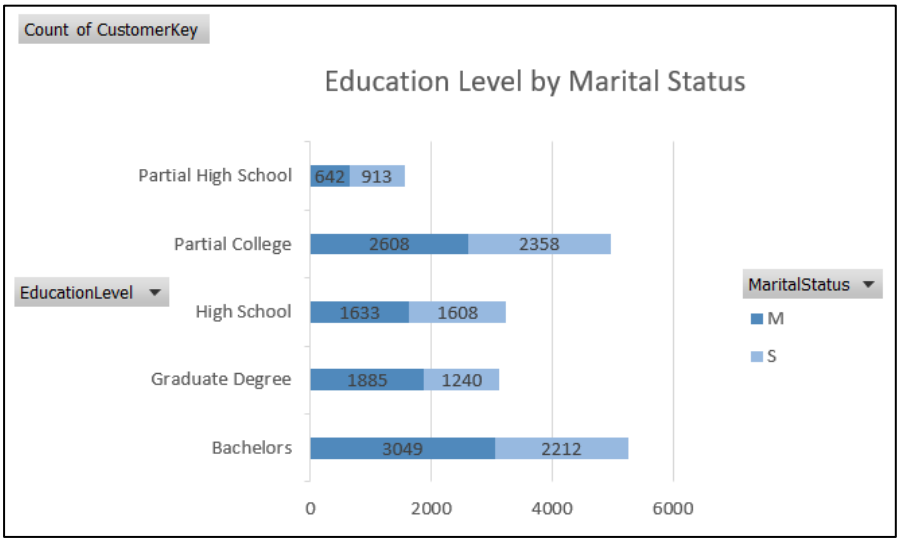
6) Home Owners vs Non-Home Owners(Doughnut Chart)
Axis (Categories): HomeOwner
Values: Count of CustomerKey



Explanation: A ring-shaped chart that compares people who own homes to those who don't.

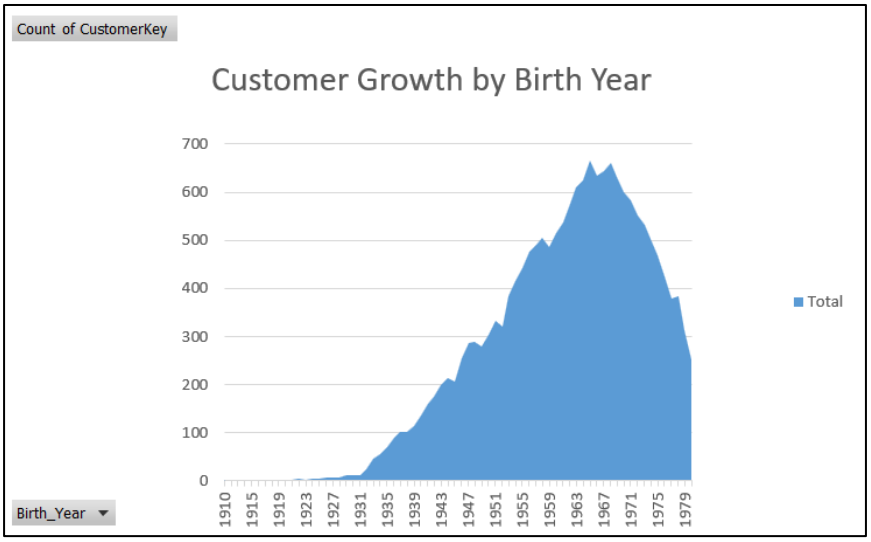
7) Education Level by Marital Status(Stacked Bar)

Legend (Series): MartialStatus
Axis (Categories): EducationalLevel
Values: Count of CustomerKey



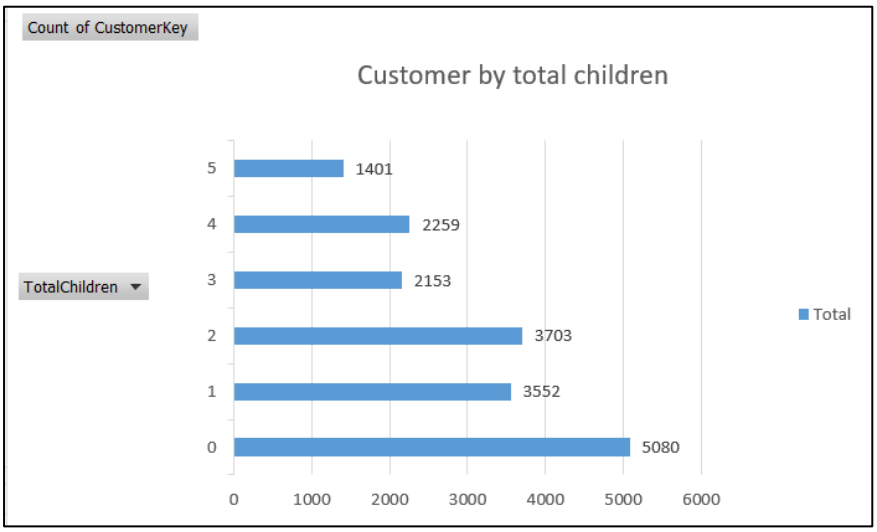
Explanation: A side-by-side view of how marriage status changes with education levels.

8) Customer Growth by Birth Year(Area Chart)
Axis (Categories): Birth_Year
Values: Count of CustomerKey



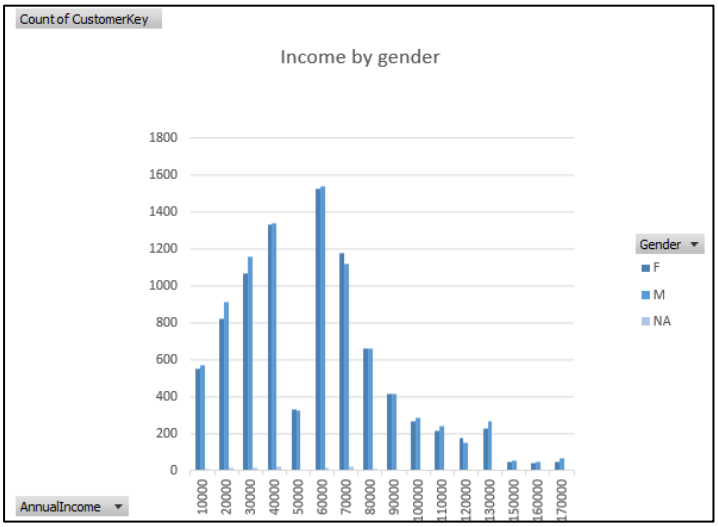
Explanation: A shaded chart that shows the "mountain" of customers across different years.

9) Customer by total children(bar chart)
Axis (Categories): TotalChildren
Values: Count of CustomerKey



Explanation: A simple chart showing which family size is the most popular among customers.

10) Income by gender(clustered column chart)
Legend (Series): Gender
Axis (Categories): AnnualIncome
Values: Count of CustomerKey



Explanation: A chart that puts male and female income bars next to each other for easy comparison.